A robot that acts to know

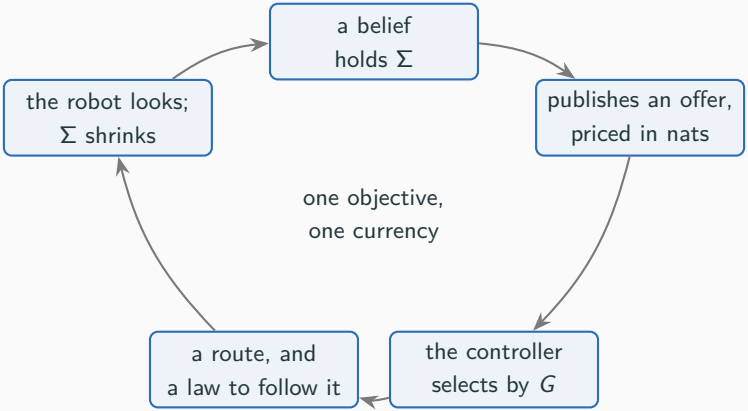
Day 4 — Action

Course built on the doctoral thesis of Noé Zapata

Thursday 3 December

Montevideo, 30 November – 4 December 2026

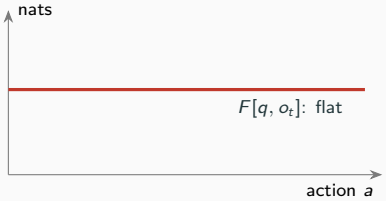
Today: from a belief to a motion, and back



And at the end the robot turns the same objective on its own body.

Why action needs a second functional

Monday's frame, now done properly

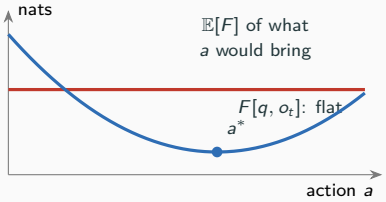


Perception. o_t is fixed, so only q can move.

Action. At decision time o_t is a number in a buffer:

$$\frac{\partial F[q, o_t]}{\partial a} = 0.$$

Monday's frame, now done properly



Perception. o_t is fixed, so only q can move.

Action. At decision time o_t is a number in a buffer:

$$\frac{\partial F[q, o_t]}{\partial a} = 0.$$

What action *can* change is the free energy of data not yet received:

$$a^* = \arg\min_a \mathbb{E}_{q(o_{t+1}, s_{t+1} \mid a)} \big[F[q, o_{t+1}] \big]$$

source: active_inference.tex l.335–369

Expected free energy

For a policy π, with predicted states and observations $\tilde{s}, \tilde{o}$, and goals entering as a **prior over preferred outcomes** $p(\tilde{o})$:

$$G(\pi) = \mathbb{E}_{q(\tilde{s},\tilde{o}\,|\,\pi)}\big[\ln q(\tilde{s}\,|\,\pi) - \ln p(\tilde{o},\tilde{s})\big]$$

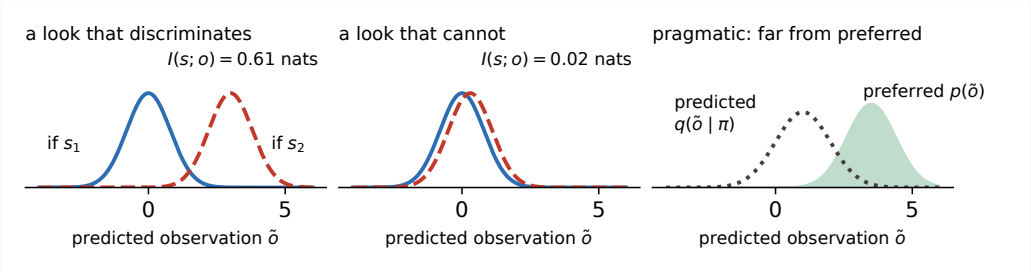
Add and subtract $\ln q(\tilde{s}\,|\,\tilde{o},\pi)$, and split the joint. Exactly:

$$G(\pi) = \mathbb{E}\Big[\underbrace{\ln q(\tilde{s}\,|\,\pi) - \ln q(\tilde{s}\,|\,\tilde{o},\pi)}_{\text{epistemic}} \underbrace{-\ln p(\tilde{o})}_{\text{pragmatic}} + \underbrace{\ln q(\tilde{s}\,|\,\tilde{o},\pi) - \ln p(\tilde{s}\,|\,\tilde{o})}_{\to 0 \text{ if inference stays good}}\Big]$$

No reward channel, no separate exploration bonus: both come out of the same line.

source: active_inference.tex l.371–412

Two terms, one currency



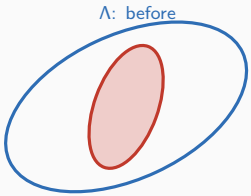
Epistemic $= -I(\tilde{s}; \tilde{o} \mid \pi)$: largest when what you would see depends on what is true.

Pragmatic $= -\mathbb{E} \ln p(\tilde{o})$: what a cost function would have held.

Both are in nats, derived together. The balance moves as the beliefs sharpen.

source: active_inference.tex l.414–427; curves: figures/make_day4.py, block 1

For Gaussian beliefs, three names for one number



ΔH is the log of the area ratio.

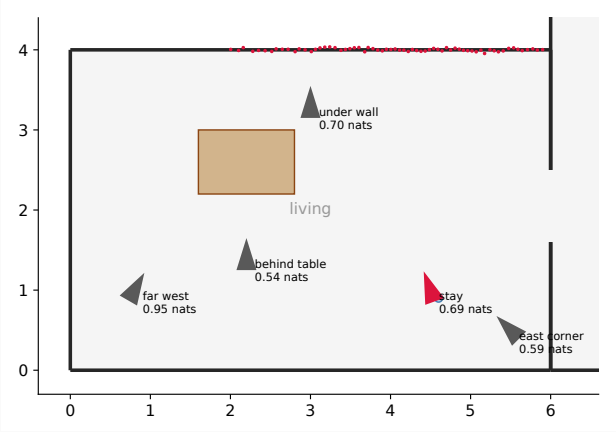
$$\underbrace{H[q] - \mathbb{E}H[q \mid o]}_{\text{expected entropy drop}} = \underbrace{I(s; o)}_{\text{mutual information}}$$

$$= \underbrace{\tfrac{1}{2} \ln \frac{|\Lambda + J^\top R^{-1} J|}{|\Lambda|}}_{\text{D-optimal design}}$$

The epistemic term, the information gain of active sensing and the criterion of optimal experiment design are one quantity. It depends on *where* you look, not on *what* you will see, so it can be computed before moving.

source: appendix_math.tex l.742–779, l.1236–1287

Where to look next

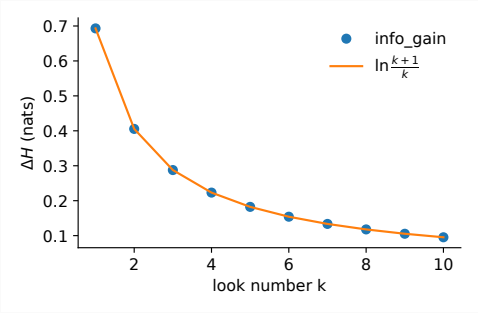


Monday's wall: the robot (red) has seen its east part. Five candidate viewpoints.

- "far west" sees few points and gains the most: it adds lever arm where the information is weak;
- "stay" gains exactly $\ln 2$. Why?

source: lab 4, exercise 1

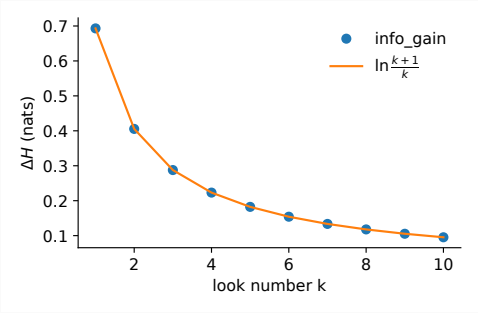
Diminishing returns, and a warning



The k-th identical look gives
$|(k{+}1)\Lambda_1|/|k\Lambda_1| = ((k{+}1)/k)^2$ for two parameters, so

$$\Delta H_k = \ln \frac{k+1}{k}.$$

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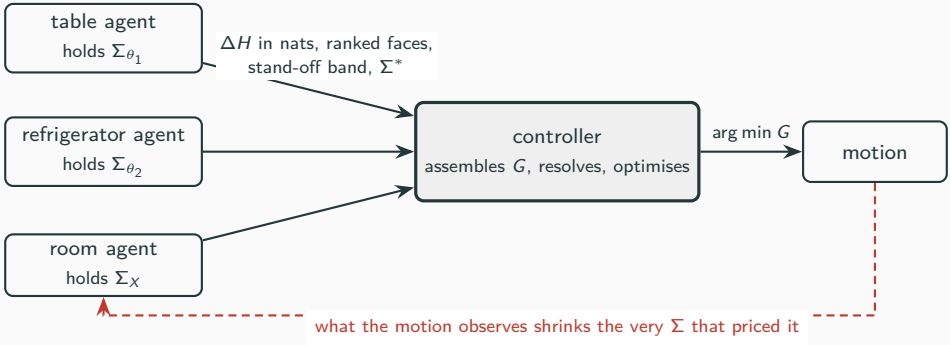
But this assumes each look's errors are independent. Repeated looks share the pose and the calibration, and those do not average away.

Day 3's common-mode ceiling already put a number on how little a repeat look is worth.

source: lab 4, exercise 2

Affordances: published, priced offers

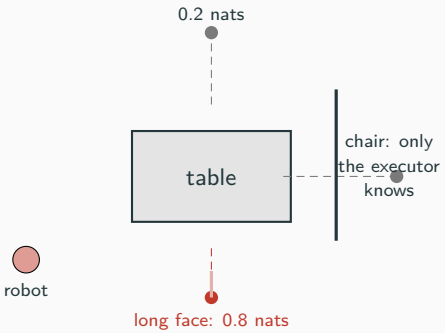
Each term of G belongs to the process that can compute it



An offer is a node in the graph. Add a class of object and you add terms to G; the controller does not change.

source: action.tex l.78–84, l.146–188, l.1478–1481

The split of knowledge



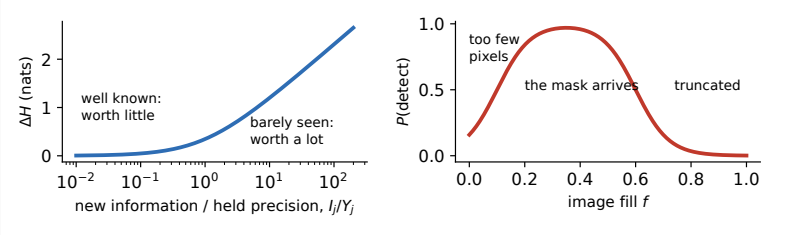
producer	which view, *relative to its object*; what it is worth; when it has learnt enough (Σ^*)
executor	feasibility; the cost of getting there; how to move and stop

A producer cannot publish a world pose. A demand that may be impossible has no vocabulary for saying so; a constraint can be negotiated.

source: action.tex l.42–45, l.109–122

The price of a view

$$\mathbb{E}[\Delta H](\mathbf{v}) \;=\; P(\text{detect}\,|\,\mathbf{v}) \sum_j \tfrac{1}{2} \ln\!\left(1 + \frac{I_j(\mathbf{v})}{Y_j}\right)$$

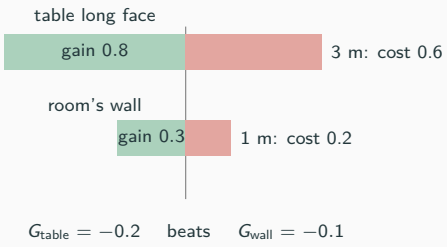


$f_{\min}$=0.10, $f_{\max}$=0.60, w_f=0.06 are a prior, not a measurement. And because the band is unimodal, a robot unsure of an object's size stands further back: no margin was set.

source: action.tex l.193–207, l.218–226, l.256–265

Selecting: one comparison, every cycle

$$a^\star = \arg\min_a \left[\lambda \cdot \mathrm{dist}(a) - \Delta H(a) - \mathrm{pragmatic}(a)\right], \qquad \lambda = 0.2 \text{ nats/m}$$



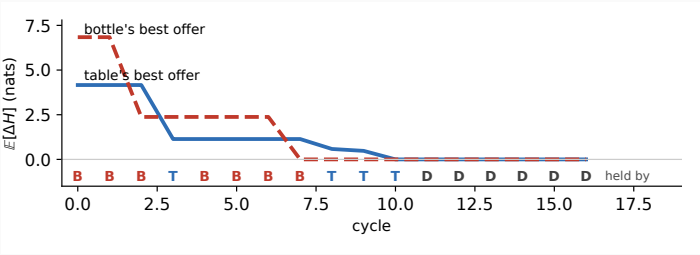
Illustrative gains.

Already driving to the wall? The table must win by a **switching margin**, 0.5 nats: the price of work already invested.

- It is inside G, not a gate on it.
- The offer in hand is re-scored with the rest.

source: action.tex l.42–49, l.329–347

An offer loses value by succeeding

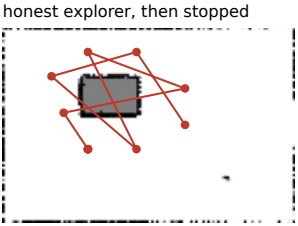


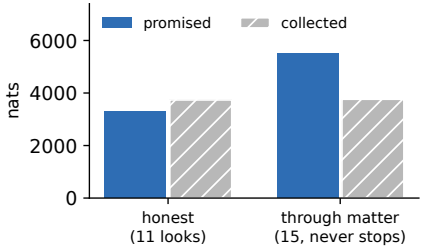
The toy of today's lab: two producers and a door worth 1 nat of preference. Nobody wrote "the table is done, go to the bottle". Each look shrinks the covariance that priced the look, until another offer is better by more than the margin.

Interruption is a consequence of the values, not a rule in the executor.

source: action.tex l.347–353; lab 4, exercise 7

The room: predict through everything you believe





An explorer that casts its predicted beams through what it already knows is solid promises cells nobody can reach, collects nothing for them, and offers them again at the next decision: the robot would shuttle between two standpoints. Predicted against realised gain is the most useful diagnostic an explorer has.

source: layout_acquisition.tex l.725–728; lab 4, exercises 8–10

What is not established

Measured: derived objective against a heuristic

The explorer written in nats, no free weights, against the hand-tuned scorer, over a population of simulated rooms: **indistinguishable in the mean**. It made a majority of rooms slightly worse and rescued a few outright failures.

The explorer running now scores unseen floor per metre: a coverage rule.

Still design, not mechanism:

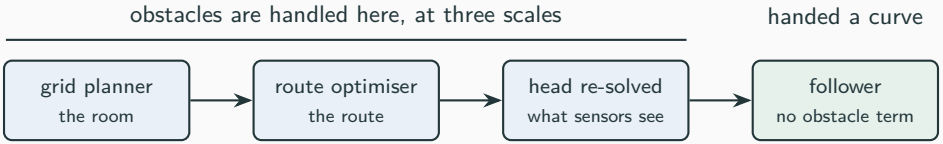
- the horizon is one affordance deep: the next look, not the best order;
- the pragmatic channel is thin;
- an offer completes on a detection, not on reaching Σ^*.

The case for the derived form rests on its having no free parameters, not on performance.

source: active_inference.tex l.876–897; action.tex l.1420–1436

From a choice to a motion

Who answers for obstacles? The geometry



The follower has no avoidance mode and no arbitration between avoiding and tracking. An arbitration problem has become a **coverage requirement**.

Others choose differently: timed elastic bands emit the commands themselves; predictive contouring keeps obstacles in the controller; regulated pure pursuit carries a collision check, the price of replanning at 1 Hz and tracking at 20.

source: action.tex l.985–1008

Routes as optimisation

$$J[r] = w_c \underbrace{\left\langle u^2 + 4u^4 \right\rangle}_{\text{clearance}} + w_\kappa \underbrace{\left\langle (\rho\kappa)^2 \right\rangle}_{\text{bending}} + \lambda_a \underbrace{\sum_k \mathrm{Huber}(\delta_k)}_{\text{authored waypoints}} + \lambda_j \underbrace{\left\langle (\mathrm{d}\kappa/\mathrm{d}s)^2 \right\rangle}_{\text{turn-rate roughness}}$$

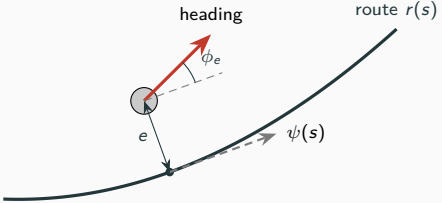
- The quartic: margin is surrendered only where it is plentiful.
- $\rho = v_{\max}^2 / a_{\mathrm{lat}}^{\max}$: bending is priced by the speed it costs.
- Terms are *not* normalised by their current size, so re-optimising an optimal route leaves it alone.

fixed tour	before	after
5th-pct. clearance (m)	0.334	0.431
lap time (s)	131.3	116.8
waypoint fidelity (m)	0.115	0.173

The third row is the price; an earlier speed limit, before λ_j.

source: action.tex l.837–876, l.878–904

The error, in the curve's own coordinates



$$\dot{e} \approx v\,\phi_e, \qquad \dot{\phi}_e = \omega - v\,\kappa(s)$$

Heading error turns forward motion into sideways drift. Heading error grows by whatever turn rate exceeds what the curve demands.

Which suggests the law:

$$\omega = \underbrace{v\kappa(s)}_{\text{feedforward}} - v\left(\frac{2}{L}\,\phi_e + \frac{1}{L^2}\,e\right)$$

source: action.tex l.1019–1027, l.1069–1077

A law you can write down

Measured: the earlier follower was a sampler

An offline replay found the lateral term contributing **91.6%** of the sampler's cost: in effect a lateral regulator paying for K rollouts. The tracker replaced it, and the sampler is gone from the build.

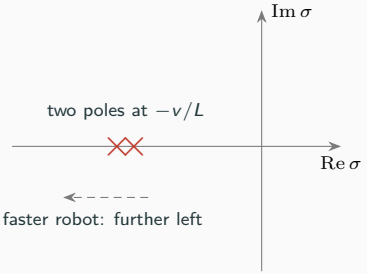
The feedforward is exact. The unicycle is differentially flat with respect to its position, so a C^2 curve and a speed profile fix $\omega = v\kappa$ in closed form.

A sampler earns its cost with multimodal costs and obstacles inside the objective. Here the obstacles live in the route.

When the dynamics are locally linear and one term dominates, write the law.

source: action.tex l.1040–1046, l.1054–1062

Pole assignment: the gains are not tuned



Substituting the law into the error equations, at constant speed:

$$\ddot{e} + \frac{2v}{L}\,\dot{e} + \frac{v^2}{L^2}\,e = 0 \implies (\sigma + v/L)^2 = 0$$

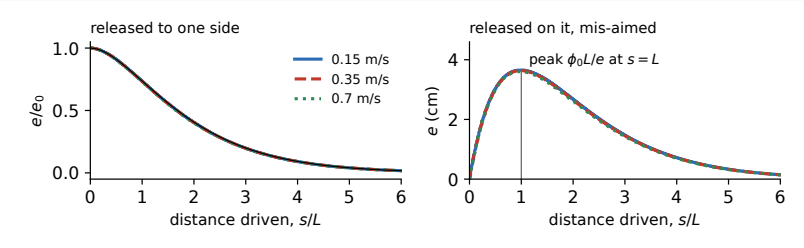
Critically damped by construction, with one free parameter, because a double pole fixes both gains:

$$k_d = \frac{2}{L}, \qquad k_p = \frac{v}{L^2}.$$

$L = 0.50$ m. What was chosen is where the poles sit.

source: action.tex l.1100–1109, l.1163–1181

The error decays over a distance, not a time

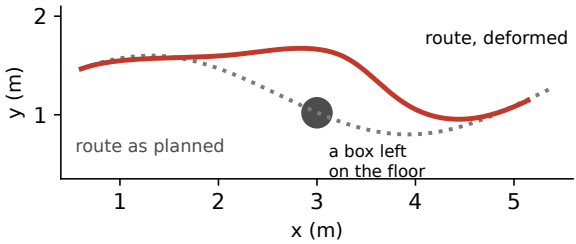


With $\mathrm{d}/\mathrm{d}t = v\,\mathrm{d}/\mathrm{d}s$ the speed cancels: $e(s) = (A + Bs)\,e^{-s/L}$. A robot crawling through a door and one crossing a room converge over the same metres, with no gain schedule. These are the toy's three speeds; thin black is the formula.

L is not a gain. It is a length, read off the space the robot works in.

source: action.tex l.1185–1198, l.1229–1236, l.1257–1260; lab 4, exercise 12

Why this follower needs no obstacle term



Same follower, unchanged. Only the curve moved: clearance −40 cm straight through, +23 cm after deforming.

What it costs. An obstacle outside the live field, or inside the short fixed prefix, is not avoided at all. No comparison with the window disabled has been run: the window is justified by the division of labour, not by a measured benefit.

source: action.tex l.1002–1008; lab 4, exercise 14

Following and choosing are one objective, at two depths

$$\nabla_{u_t} G_T(\pi) = \underbrace{\nabla_{u_t} G_{t+1}(u_t)}_{\text{what the follower handles, in closed form}} + \underbrace{\sum_{\tau=t+2}^{t+T} \nabla_{u_t} G_\tau(\pi)}_{\text{what the selector adds, in affordances}}$$

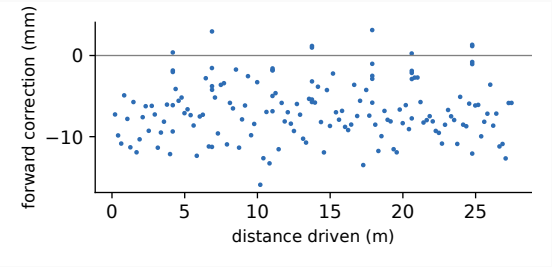
keep a re-evaluated **goal** from dithering: switching margin, in nats ⟷ same device ⟷ keep a re-solved **route** from jumping: proximal prior, 3 cm

MPC, MPPI and planning as inference sit on this line, and differ in how many terms they keep and how they estimate them.

source: active_inference.tex l.504–508, l.554–579; action.tex l.978, l.1396–1408

The same objective, turned on itself

Calibration is inference, not maintenance



The toy's wheels over-report by 4%. The room corrects each step, and every correction points the same way.

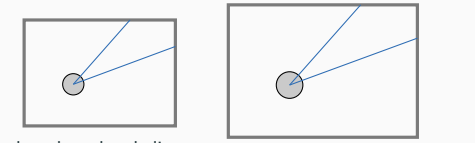
A wider covariance would not help: it says how far the prediction may be wrong, not in *which direction*.

Put (s_v, s_ω, b) in the state: NEES 8.3 $\rightarrow$ 1.6.

If the body is a factor of the posterior, estimating it is the same inference.

source: active_inference.tex l.797–800; self_calibration.tex l.53; lab 4, exercises 15–17

What no data can determine



what the robot believes

everything 2% larger:
wheels, track, ranges, room, mounts

No reading changes.

A transformation that changes no factor of the model is a **gauge**. The data cannot fix it; only a stated convention can.

- Seven survive in the joint model.
- Scale: trust the LiDAR, $s_L \equiv 1$. Every scale reported is relative to it.
- The information computation finds this one: 5×10^{-7} along its direction.

source: self_calibration.tex l.643–700

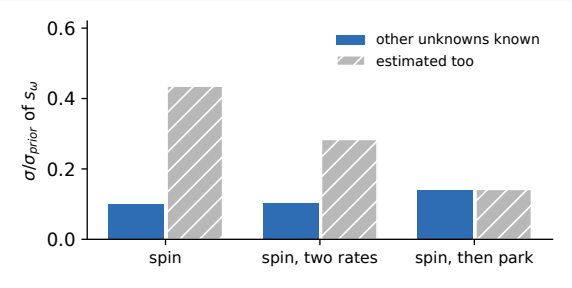
What each motion makes observable

	k_r	δ	k_b	k_ω	b_ω	x_h	y_h	ε_{yaw}	τ_o	τ_g
parked	–	–	–	–	0.59	–	–	–	–	–
straight	0.23	0.05	–	–	0.59	–	–	0.26	0.12	–
spin, one rate	0.71	–	0.71	0.05	–	0.08	0.08	–	–	–
spin, two rates	0.71	–	0.71	0.04	0.86	0.08	0.08	–	0.09	0.06
tour	0.34	0.08	0.35	0.05	0.59	0.11	0.11	0.41	0.04	0.02

Posterior over prior sd, every other parameter estimated too; "–" means the motion teaches nothing.
Standing still teaches the gyro bias and nothing else. Straight and spin teach disjoint sets.
Observability is a property of the parameter and the motion together.

source: self_calibration.tex l.868–877, tab:sc-obs-motion; a subset of the rows

Conditional is not marginal



At a constant turn rate, scale and bias both grow in proportion to time: the data see one combination.

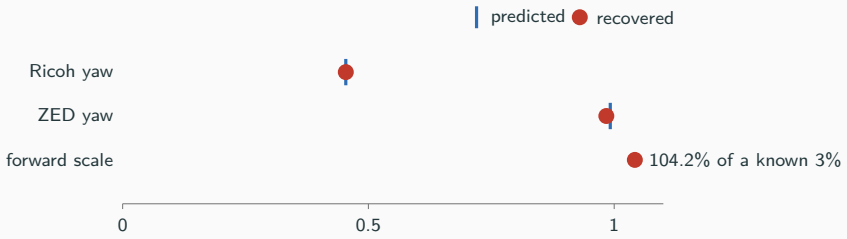
$$S = \mathcal{I}_{\theta\theta} - \mathcal{I}_{\theta n}\,\mathcal{I}_{nn}^{-1}\,\mathcal{I}_{n\theta}$$

The Schur complement removes what another unknown could equally have produced. A second rate, or a pause, splits them.

source: self_calibration.tex l.820–830; lab 4, exercise 18

Testing it: inject an error, and predict the recovery first

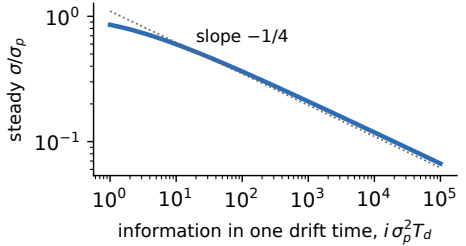
$$\text{recovery} \;=\; 1 - \left(\frac{\sigma_{\text{post}}}{\sigma_{\text{prior}}}\right)^2$$

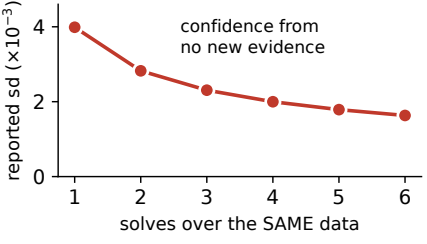


Run twice with the same seed: the difference sees only the injection. It is predictable from the posterior alone, so running both checks that the estimator is consistent with itself.

source: self_calibration.tex l.3056, l.3640–3644

A body that drifts, and the prior that must not move





Drift. A body that drifts a thousand times more slowly is tracked only about six times more tightly.

Never re-centre a prior on its own estimate. The same evidence counts again, and a parameter nothing excites wanders. Decided on the argument: no measurement yet isolates the effect.

source: self_calibration.tex l.1001–1017, l.2763–2825; right panel: lab 4, exercise 21

Standing still is an observation

parameter	information per second
k_v forward scale	$v^2 T/\varsigma_p^2$
k_ω heading scale	$\omega^2 T/\varsigma_\theta^2$
b_ω gyro bias	T/ς_θ^2

No rate in the last row: only time. Two legs of one route, differing in whether the robot happened to stand still: the data's share of the bias posterior goes from **about half** to **almost all**.

The world is sampled only where the robot looks. The body is in everything it does.

So ordinary driving calibrates, and a manoeuvre earns its place only where the task does not span the covariate; a planner avoids a motion exactly as much as the motion costs it. It should compete with idleness, never with a task.

Here the one calibration action the analysis justifies is a pause.

source: self_calibration.tex l.2670–2762; active_inference.tex l.852–868, l.957–960

Lab

Lab 4: paying for what you learn

`labs/day4.ipynb`, about two hours.

1. **Where to look**: ΔH on one wall, and travel priced in nats.
2. **Priced offers**: producers publish on the graph, the controller selects; find the hand-offs nobody coded.
3. **The room**: an explorer that knows when to stop, and the trap of predicting through matter.
4. **Following**: a double pole, three ways to break it, a route deformed around an obstacle.
5. **The body**: recover hidden wheel and gyro errors from the room's corrections; what each motion teaches.

Short of time: skip exercises 2, 16, 19 and 20.

Tomorrow

The programme for Day 5 is to be confirmed.

Today's result to carry there: what can be learnt is a property of the parameter and the action together. A robot pays for what it knows; the question left is how anyone can tell whether it paid well.